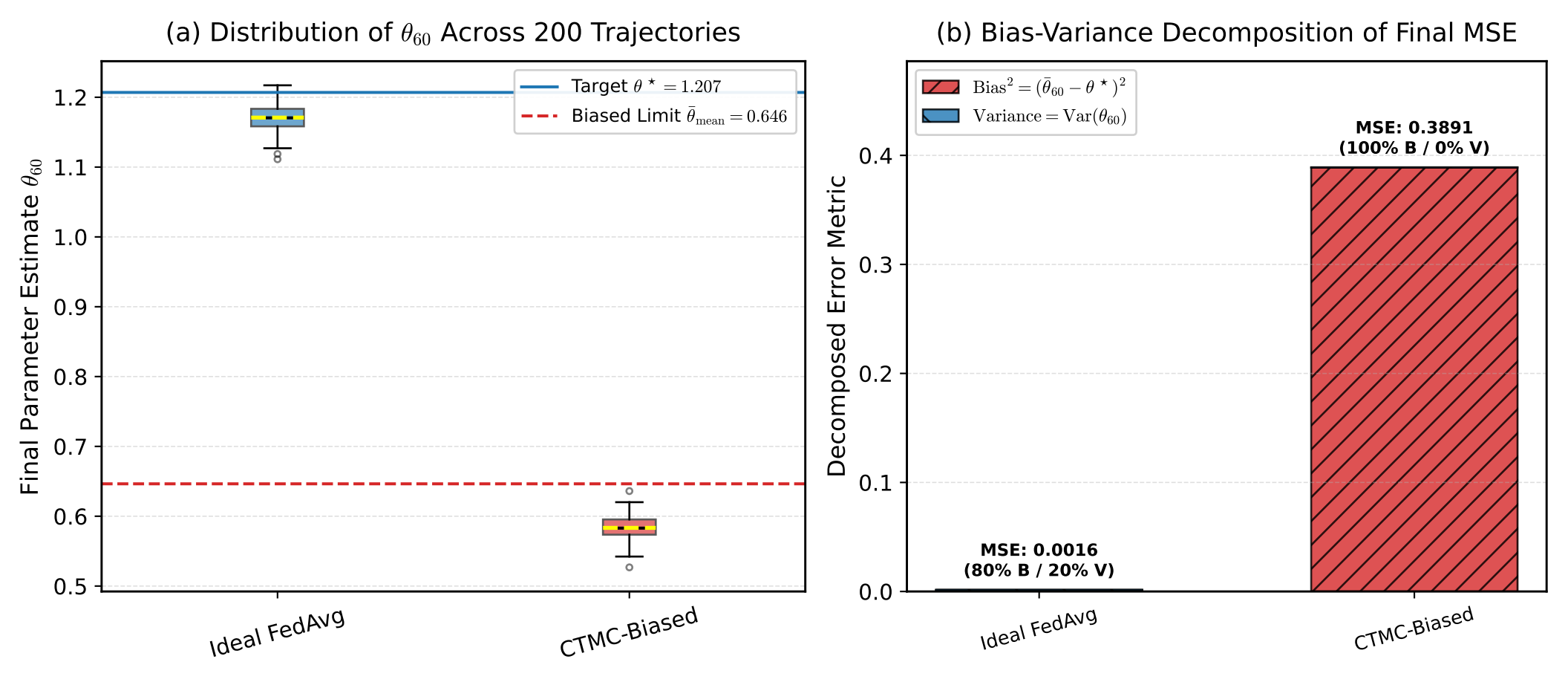
(a) Distribution of θ_{60} Across 200 Trajectories
Final Parameter Estimate θ_{60}
Target $\theta^\star = 1.207$
Biased Limit $\bar{\theta}_{\text{mean}} = 0.646$
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
Ideal FedAvg
CTMC-Biased
(b) Bias-Variance Decomposition of Final MSE
Decomposed Error Metric
$\text{Bias}^2 = (\bar{\theta}_{60} - \theta^\star)^2$
$\text{Variance} = \text{Var}(\theta_{60})$
MSE: 0.0016
(80% B / 20% V)
MSE: 0.3891
(100% B / 0% V)
0.4
0.3
0.2
0.1
0.0
Ideal FedAvg
CTMC-Biased